

# Improvement of Parkinson's Disease Motor Symptoms and Alpha-Synuclein Pathology via CJRB-302: A Live Biotherapeutic Product from a Healthy Human Gut Microbiota

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**Introduction.** Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by motor symptoms with various non-motor symptoms<sup>1,2</sup>. Pathological hallmarks include Lewy bodies and loss of dopaminergic neurons in the substantia nigra pars compacta (SNpc)<sup>1,2</sup>.

In recent years, accumulating evidence suggests that the microbiota-gut-brain axis, a complex bidirectional pathway, plays a role in PD pathogenesis. Altered gut microbiota in patients with PD have been observed, including an increase in *Bifidobacterium*, *Lactobacillus*, and *Akkermansia*, and a decrease in *Prevotella*, *Roseburia*, *Blautia*, *Lachnospira*, and *Faecalibacterium*<sup>3,4</sup>.

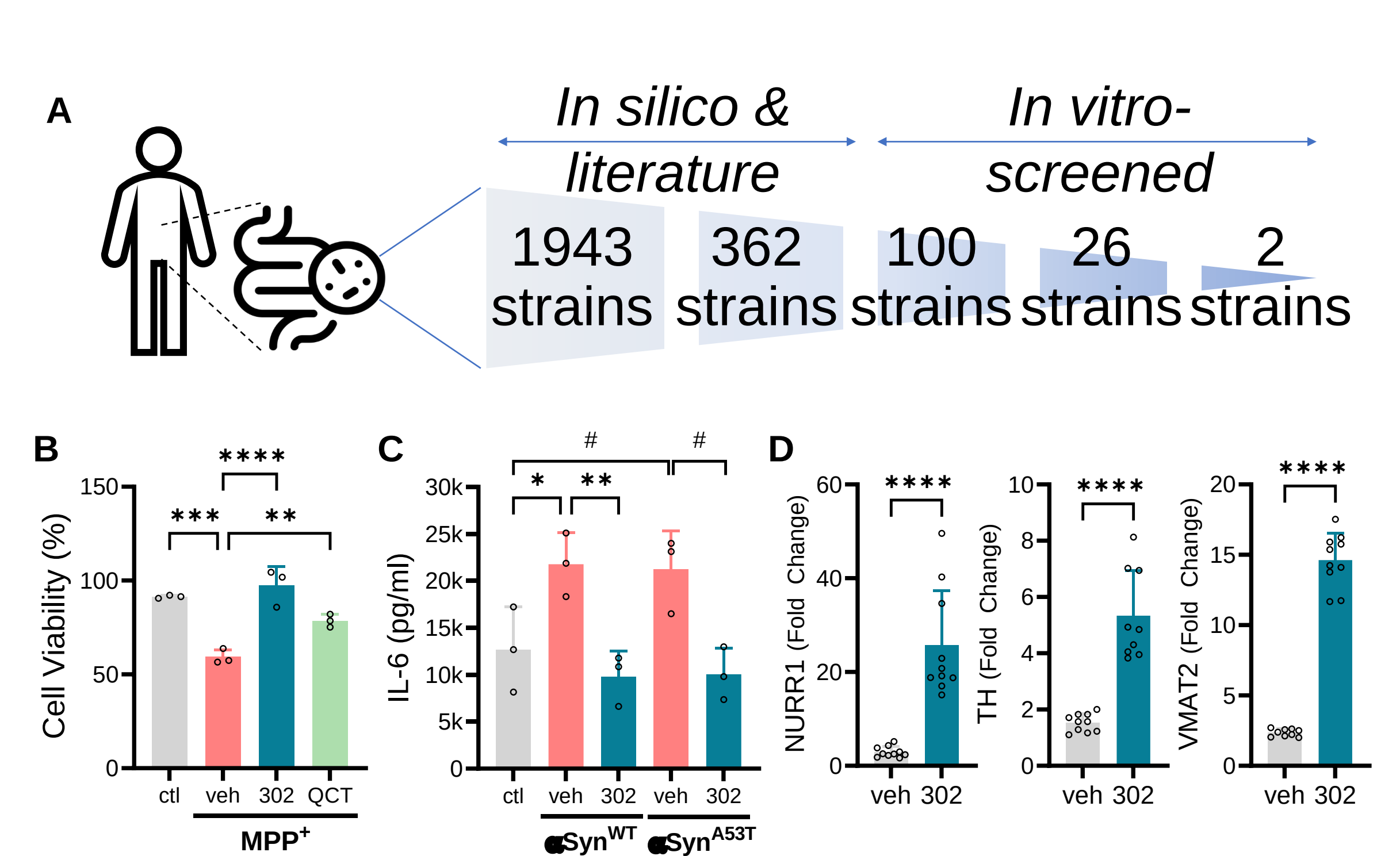
In preclinical studies, fecal microbiota transplantation (FMT) from patients with PD to alpha-synuclein (aSyn) overexpressing (ASO) mice exacerbates motor deficits and aSyn pathology than wild-type mice<sup>5</sup>. Clinical FMT studies have shown a 5 to 13-point improvement in UPDRS motor scores, highlighting the microbiota-gut-brain axis' critical role in PD<sup>6-8</sup>. However, FMT has disadvantages like the pathogen transfer potential and low reproducibility.

Given that, there is a definite need to develop controlled and reproducible microbiome-based therapies for PD. In this study, we investigated the therapeutic potential of CJRB-302, a live biotherapeutic product (LBP) identified from healthy human gut microbiota.

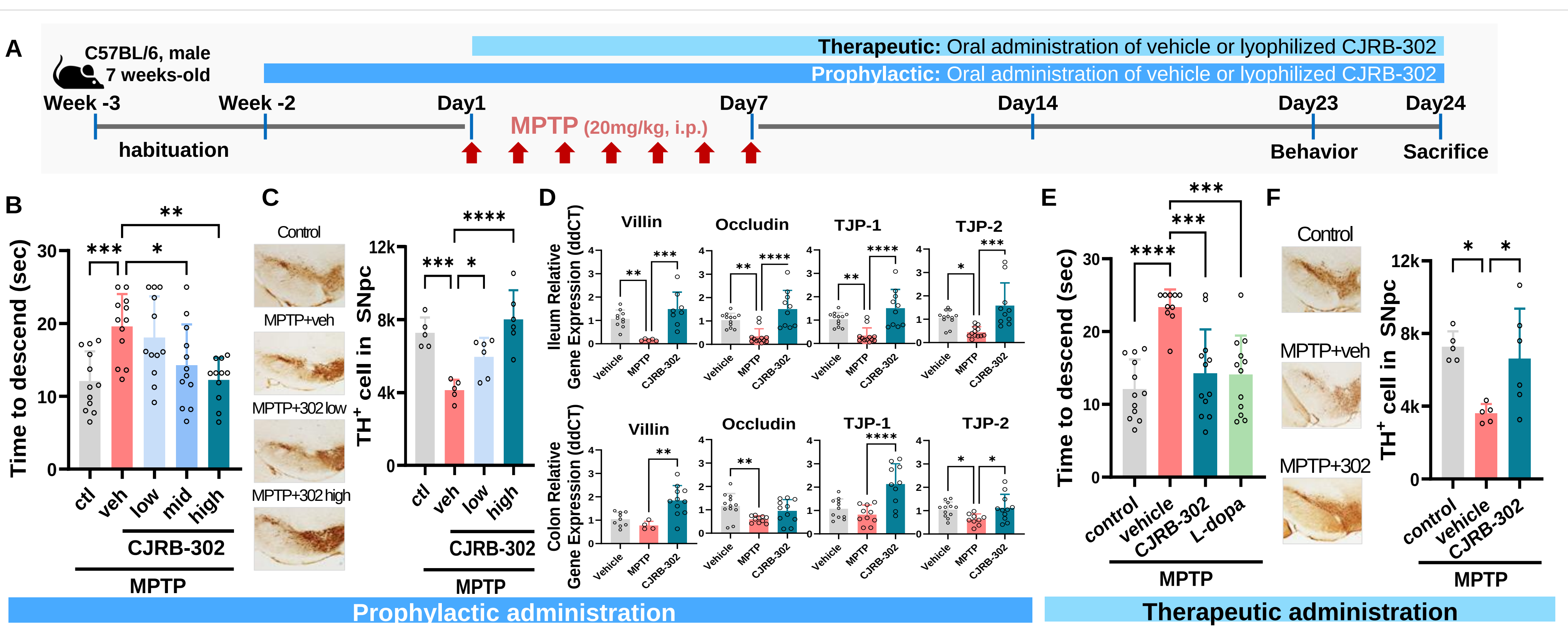
**Methods. Screening & in vitro studies.** CJRB-302 was selected from 1,943 gut-derived strains through screening and evaluated for antioxidant, anti-inflammatory, and neuroprotective effects *in vitro*. **Drug formula:** CJRB-302 was cultured, harvested by centrifugation, mixed with a cryoprotectant, lyophilized, and aliquoted into airtight glass vials, all under anaerobic conditions in a semi-GMP facility. The vials were stored at -20°C until administration, then suspended in anaerobically equilibrated PBS to 10<sup>5</sup>, 10<sup>6</sup>, or 10<sup>7</sup> CFU/mouse/day and administered via oral gavage. **In vivo studies:** Mice injected with 1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) were treated with CJRB-302. Motor deficits and tyrosine hydroxylase (TH)+ cell counts were assessed. Fecal samples were collected and analyzed at the end of the study. Mice with unilateral AAV-aSyn-A53T injection were evaluated for motor deficits, TH+, and pS129+ cell counts. Liver toxicity was also assessed.

**Findings. In vitro.** CJRB-302 reduced aSyn-induced inflammatory responses and MPP+-induced cytotoxicity in differentiated SH-SY5Y cells, while increasing dopaminergic markers including TH and VMAT2. In addition, based on gut organoid studies, CJRB-302 increased tight junction-related gene expression. **MPTP mouse model.** In mice, orally administered CJRB-302 significantly alleviated dopaminergic cell loss in the SNpc and motor deficits caused by MPTP injection. Accordingly, CJRB-302 shifted the gut microbiota towards a profile observed in healthy individuals, correlating with improved neuronal pathology, motor function, and intestinal Nurr1 protein levels. Also, CJRB-302 increased tight junction-related gene expression in the gut. **AAV-aSyn model.** Notably, in AAV-aSyn-induced mice, oral CJRB-302 reduced aSyn pathology, motor deficit, and dopaminergic cell loss in the SNpc without exhibiting toxicity. Ongoing research explores the mechanisms behind CJRB-302's beneficial effects to elucidate its mode of action in PD pathogenesis. Moreover, clinical trials are imperative to establish the efficacy and safety of CJRB-302 in human PD patients.

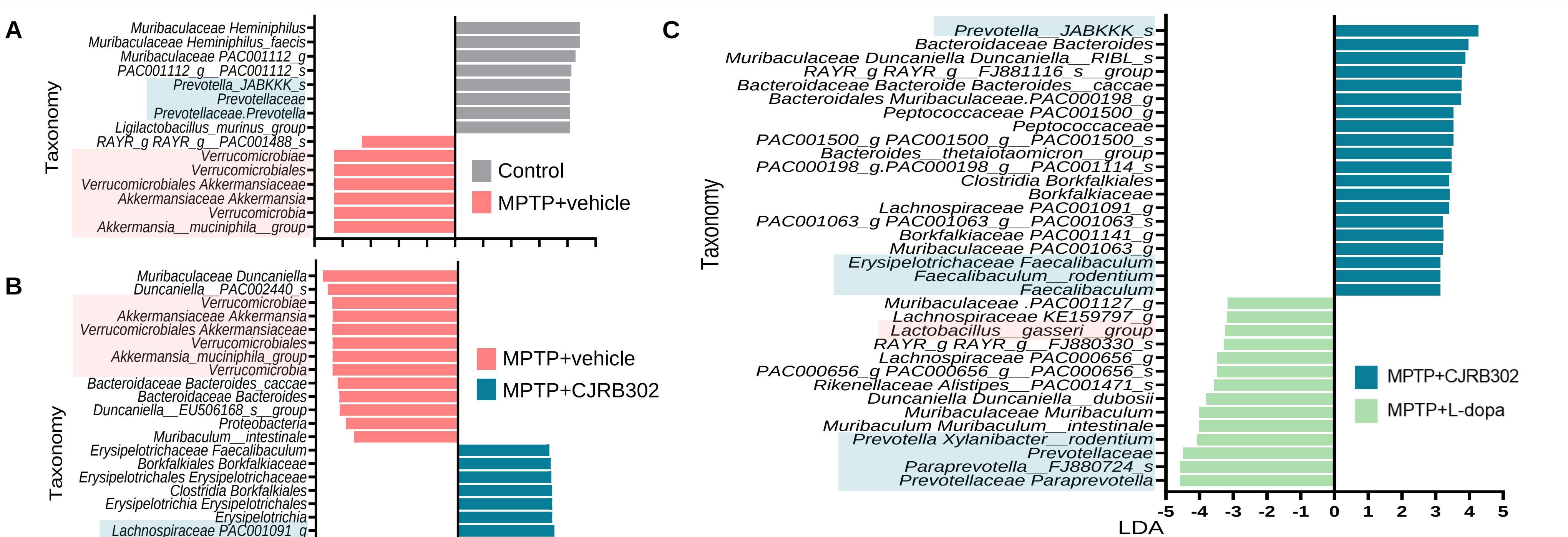
**Conclusion.** This study supports CJRB-302 as a promising new drug candidate for the development of innovative PD therapies.



**Figure 1. Selection of CJRB-302 as Parkinson's disease pipeline** (A) Screening procedures in CJRB-302 selection. (B) Neuroprotective effects of CJRB-302 supernatants in MPP+-induced cytotoxicity in differentiated SH-SY5Y. (C) Effects of CJRB-302 supernatants in aSyn-induced IL-6 secretion in U-373 and SH-SY5Y co-culture. (D) Upregulated expression of Nurr1 and target gene transcripts in SH-SY5Y. Error bars indicate the mean ± S.D. \*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001, and \*\*\*\*p < 0.0001 determined by one-way ANOVA with Dunnett's post hoc analysis for (B) and (C), unpaired student's t-test for (D).

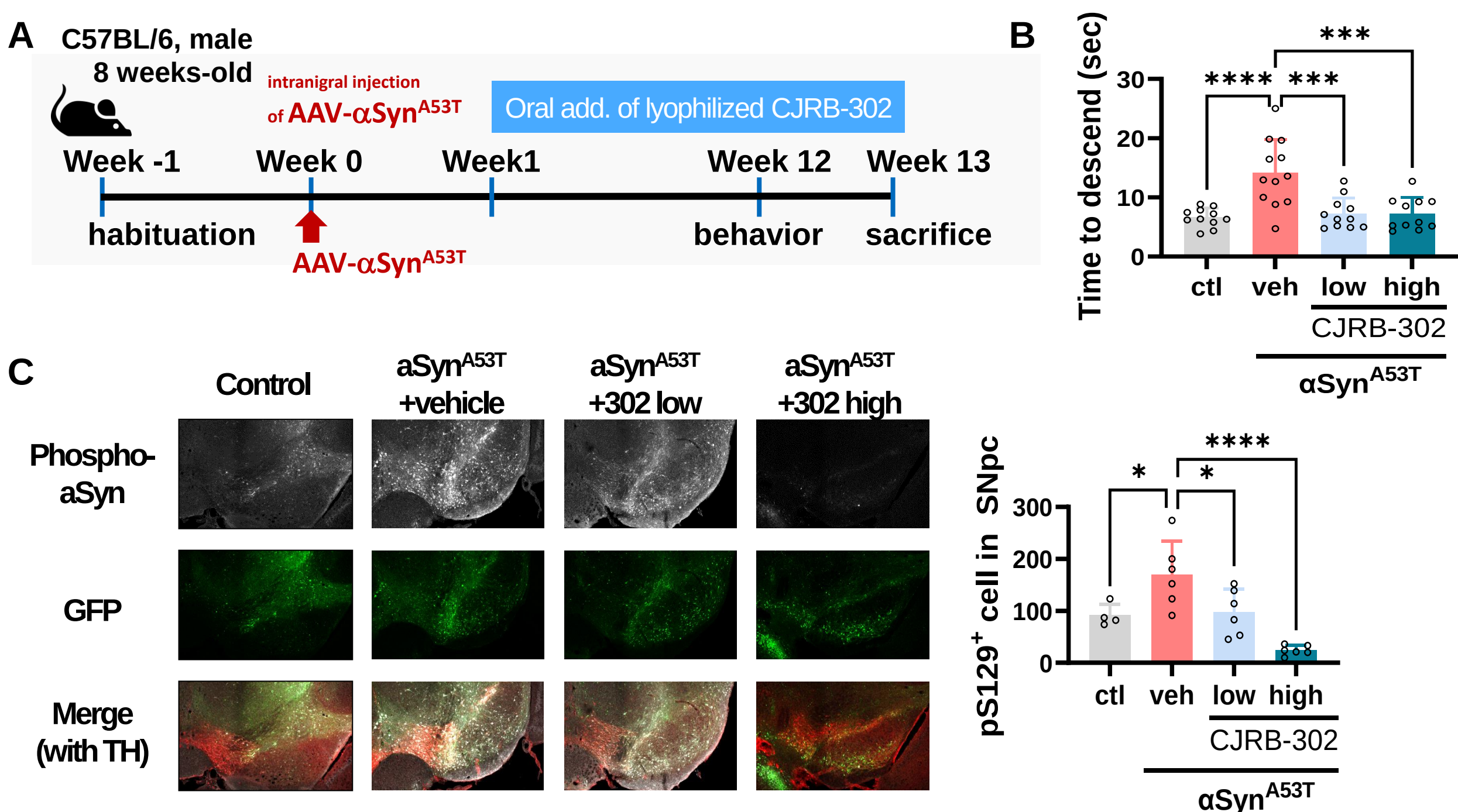


**Figure 2. Oral administration of CJRB-302 ameliorates motor deficits and neuronal loss in MPTP-induced PD mouse model.** (A) Schematic illustration of CJRB-302 administration in MPTP-induced PD mouse model. (B-C) CJRB-302 administered group displayed enhanced performance in pole test (B) and higher nigral TH+ cell number (C) in prophylactic administration schedule. (D-F) CJRB-302 administered group displayed enhanced performance in pole test (D), rotarod test (E), and higher nigral TH+ cell number (F) in therapeutic administration schedule. Error bars indicate the mean ± S.D. \*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001, and \*\*\*\*p < 0.0001 determined by one-way ANOVA with Dunnett's post hoc analysis.



**Figure 3. Oral administration of CJRB-302 to MPTP-injected mice altered PD-patient-like microbiota composition by increasing bacteria levels that are decreased in patients with PD.** Linear Discriminant Analysis Effect Size (LEfSe) identifies differences in abundant taxa in the feces. The bar plot represents the log-fold differences between treatment groups, based on effect size.

| Table 1. Summary of Spearman correlations |                                               |               |             |
|-------------------------------------------|-----------------------------------------------|---------------|-------------|
| Metadata                                  | Taxonomy                                      | SpearmanCoef. | p-value     |
| TH+ cell                                  | Bacteroidaceae Bacteroides caca               | -0.73868992   | 2.33917E-08 |
|                                           | Prevotellaceae Xylanibacter rodentium         | 0.590711108   | 3.81646E-05 |
|                                           | Prevotellaceae Paraprevotella FJ880724        | 0.563455436   | 0.000102237 |
|                                           | Muribaculaceae Duncaniella PAC002440          | -0.515334825  | 0.000478656 |
|                                           | Lachnospiraceae PAC001074                     | -0.490270875  | 0.000980902 |
|                                           | Lachnospiraceae KE159571                      | 0.48911302    | 0.001012639 |
|                                           | Lachnospiraceae Zhenpiania PAC001808          | 0.4736526     | 0.001533058 |
| intestinal_Nurr1                          | Unknown PAC001118                             | -0.608829216  | 0.000585874 |
|                                           | Lachnospiraceae Acetivibacter muris group     | 0.534018996   | 0.003422784 |
|                                           | Unknown PAC001762                             | 0.527626571   | 0.00390835  |
|                                           | Lachnospiraceae Anaerotruncus KB822568        | 0.504815445   | 0.006149454 |
|                                           | Rikenellaceae Alistipes PAC001060             | -0.496442255  | 0.007208127 |
| Pole_test                                 | Lachnospiraceae Duncaniella PAC002440         | -0.342808443  | 0.000672106 |
|                                           | Enterobacteriaceae Klebsiella aerogenes group | 0.313549138   | 0.001974366 |
|                                           | Lachnospiraceae Enterocloster PAC002304       | 0.305254814   | 0.00262964  |
|                                           | Bifidobacterium pseudolongum group            | -0.301516683  | 0.002984326 |
|                                           | Lachnospiraceae Agathobaculum AY239438        | 0.294790786   | 0.003732051 |



**Figure 4. CJRB-302 rescues PD pathology in aSyn-induced PD mouse model.** (A) Schematic illustration of CJRB-302 administration in aSyn-induced PD mouse model. (B-D) CJRB-302 administered group displayed enhanced performance in pole test (B), higher nigral TH+ cell number (C), significantly diminished pS129 aSyn+ cells in the SNpc (D) of aSynA53T mice. Error bars indicate the mean ± S.D. \*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001, and \*\*\*\*p < 0.0001 determined by one-way ANOVA with Dunnett's post hoc analysis for (B) and (D), and Holm-Sidak's multiple comparison test for (C).

**References** 1. Fröhlich, F. Chapter 23 – Parkinson's Disease. 2016; pp.291-296. Academic Press. ISBN 9780128015605; 2.Hijaz and Volpicelli-Daley, *Mol Neurodegener.* 2020;15(1):19; 3. Lorente-Picon and Laguna, *Biomolecules.* 2021;11(3):433; 4. Chen and Lin, *J Biomed Sci.* 2022;29(1):545. Sampson et al., *Cell.* 2016;167(6):1449-1480; 6. Segal et al., *Clin Neurol Neurosurg.* 2021;207:106791; 7. Cheng et al., *Gut. Microbes.* 2023;15(2):2284247; 8. Bruggeman et al., *EClinicalMedicine.* 2024; 71:102563

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**More Information**

